VISHWAKARMA INSTITUTE OF TECHNOLOGY, PUNE – 37
(An Autonomous Institute Affiliated to Savitribai Phule Pune University)

Linear Algebra and Differential Equations
Home Assignment-6

Instructions for solving the assignment:

i) Every student is supposed to solve 2 sub - question from each of the main questions (i.e., total 2 questions) based on the following rule:
Consider last 2 digits of your GR number and solve the corresponding sub questions.
Example: If your GR number is "12411309" then the student will solve sub-question 0 of Question A, and sub-question 9 of Question B

ii) Write the assignment in the section for Home assignments in your tutorial notebook.

iii) Get the assignment checked by your tutorial teacher on or before 30.05.2025

Q.A Find a general solution of the differential equations given below.
Find the particular integral using the method of undetermined coefficients.

0) $(D^2 + 1)y = x\cos x$
1) $(D^2 - 4D + 3)y = x^2 e^x$
2) $(D^2 + 5D + 4)y = x^2 + 6x$
3) $(D^2 + 1)y = x\sin x$
4) $(D^2 - 3D + 2)y = x^2 + \cos x$
5) $(D^2 + 6D + 9)y = e^{-x}\sin 2x$
6) $(D^3 - 4D)y = x\sin 2x$
7) $(D^2 - 2D + 1)y = x\,e^x\cos 2x$
8) $(D^3 + 3D^2 - 4)y = 6e^{-2x} + x^4$
9) $(D^2 - 4D + 4)y = xe^{2x}$

Q.B Find a general solution of the differential equations given below.
Find the particular integral using the method of variation of parameters.

0) $(D^2 + 4)y = \tan^2 2x$
1) $(D^2 + 9)y = \sec 3x$
2) $(D^2 + 4)y = \sec 2x$
3) $(D^2 - 4D + 4)y = e^{2x}\tan^2 x$
4) $(D^2 - 2D + 1)y = e^{-x}\log x$
5) $(D^2 + 3D + 2)\,y = e^{e^x} + \sin e^x$
6) $(D^2 + 4)y = 4\sec 2x\tan 2x$
7) $(D^2 + 1)y = \operatorname{cosec} x\cot x$
8) $(D^2 - 2D + 1)y = e^x\tan x$
9) $(D^2 + 1)y = \cot x$